Government of Pakistan

National Vocational and Technical Training Commission

Prime Minister's Hunarmand Pakistan Program

"Skills for All"

Course Contents / Lesson Plan

Course Title: Artificial Intelligence (Machine Learning & Deep Learning)

Duration: 3 Months

Trainer Name	
Course Title	Artificial Intelligence (Machine Learning & Deep Learning)
Objective of Course	Employable skills and hands on practice for Artificial Intelligence, including specialization in Natural Language Processing (NLP) and Microsoft Azure AI Associate The aim for the team of staff responsible for delivery of the advanced IT curriculum is to provide knowledge and develop skills related to the IT. The course will allow participants to gain a comprehensive understanding of all the aspects. It will also develop the participant's ability to act in a professional and responsible manner. Teaching staff will provide the technical knowledge and abilities required to solve tasks and problems that are goal-oriented. They will use participant-centered, practically oriented methods. They will also develop a program of practical assessment that reflects the learning outcomes stated in the curriculum. Trainees of the IT curriculum will also develop their willingness and ability as individuals to clarify issues, as well as think through and assess development opportunities. Teaching staff will also support trainees in developing characteristics such as self-reliance, reliability, responsibility, a sense of duty and a willingness and ability to criticize and accept criticism well and to adapt their future behavior accordingly. Teaching staff also use the IT curriculum to address the development of professional competence. Trainees will acquire the ability to work in a professional environment. By the end of this course, the trainees should gain the following competencies: Understanding of core concepts of artificial intelligence and machine learning State of the art machine learning techniques Hands-on exposure to exploratory data analysis Practical exposure to model design, evaluation Familiarity with tools and libraries such as scikit learn, pandas numpy, tensorflow, pytorch and keras

Learning Outcome of the Course	After taking this course, you will be familiar with the fundamentals of Artificial Intelligence. You will gain practical experience in applying AI for problem solving, and will develop a deep understanding of the core concepts by implementing solutions to real world problems. By the end of this course, the trainees should gain the following competencies: Understanding of core concepts of artificial intelligence and machine learning State of the art machine learning techniques Hands-on exposure to exploratory data analysis Practical exposure to model design, evaluation Familiarity with tools and libraries such as scikit learn, pandas numpy, tensorflow, pytorch and keras After the specialization in NLP, you will be comfortable using TensorFlow pipelines for NLP at the end of the course. Moreover, You will learn to build your own models which will extract information from textual data. You will learn text processing fundamentals, including text normalization, stemming and lemmatization. You will learn about different evaluation metrics for models trained for NLP tasks. You will learn to make a part of speech (POS) tagging model. You will learn about named entity recognition. You will learn advanced techniques including word embeddings, deep learning (DL) techniques. You will learn how to deploy a NLP model Moreover, you will learn not only all these skills but also learn to use Microsoft Azure API for Machine and Deep Learning for numerical, image and text data.
Course Execution Plan	Total Duration of Course: 3 Months
	Class Hours: 4 Hours per day, 5 Days a week
	Theory: 20% Practical: 80%
Companies Offering Jobs in the respective trade	1. Careem 2. Afiniti 3. Addo.ai 4. Arbisoft 5. I2c 6. Xavor 7. Fiverivers Technologies 8. Confiz 9. Crossover 10. NetSol 11. Research institutes 12. All Private Institutes who have an ML department
Job Opportunities	AI is the buzzword of the century, attracting attention across industries, motivating changes in products as well as services. It is the very nature of the subject that makes its applications infinite,

	in multiple domains. Whether you belong to a technical background or not, chances are that AI can make your job easier, and push it in the right direction. Dive in to develop an understanding of the core concepts, while gaining hands on experience and training from the industry's finest. Trained resources can find work as one of the following roles: • AI Associate Engineer • Machine Learning associate analyst • Assistant Data Analyst • Research Assistant
No of Students	25
Learning Place	Classroom / Lab

Scheduled Week	**Module Title**			**Learning Units**	**Remarks**
Week 1	Introduction	Day 1	Hour# 1	• Introduction to AI • Motivational Lecture (For further detail please see Page No: 3& 4)	• Task 1 • Task 2 • Task 3-25 *Details may be seen at Annexure-I*
			Hour# 2	• Course Introduction • Job market • Course Applications • Work ethics • Survey of career opportunities • Survey of industry requirements for each career path •	
			Hour# 3, 4	•Software Installation (Anaconda, VSCode, PyCharm, etc.)	
	Linux Shell Scripting Fundamentals	Day 2	Hour# 1	Introduction to Debian •Basic Commands: pwd, cd, ls, cat, sudo, man, redirection, mkdir, rm, rmdir, cp, mv	
			Hour# 2	•file, reading, cat, more, less, head, alias,	
			Hour# 3	•shutdown, restart, touch, nano, bash, sh, chmod, ps, kill, dpkg	
			Hour# 4	•Package update and upgrade•Environment	
	Python Fundamentals	Day 3	Hour # 1	Values, expressions, and statements •Numbers, Booleans, Strings •Operators, variables and keywords	
			Hour # 2,3	•String operations	

			Hour # 4	•Input and Type casting •Comments	
		Day 4	Hour # 1 & 2	Data Structures • Lists • Tuples	
			Hour # 3 & 4	• Dictionaries • Sets	
		Day 5	Hour # 1 & 2	Conditional Execution •If, elif, and else statements •Break, continue, and pass statements	
				•Nested conditionals •Conditional (Ternary) Expression	
			Hour # 3 & 4	•While, for loops and use of enumerate•Nested loops •List comprehension •Iterators and Iterables •	
Week 2	Python Fundamentals	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	• Task 26-27 • Task 49-51 *Details may be seen at Annexure-I*
			Hour# 2, 3	• Functions • Functions and variable scope • Lambda expression • Map and Filter • Inner/Nested functions	
			Hour # 4	• File Handling • Exception Handling	
	Implementation of OOP Principals in Python	Day 2	Hour# 1	• Classes and Objects • Instance Variables and Methods • Class Variables and Functions • Constructors and Destructors	
			Hour# 2,3	• Inheritance • Multilevel Inheritance • Hierarchical Inheritance • Multiple Inheritance, Method Resolution Order	
			Hour# 4	• Access Specifiers: Private, Public, Protected • Name Mangling • Inner/Nested Class • Association, Aggregation, Composition	
		Day 3	Hour# 1	• Polymorphism and Operator Overloading	
			Hour# 2	• Magic Functions/Dunder Functions	
			Hour# 3	• Dynamic Polymorphism (subclass as base class)	

			Hour# 4	• Abstract Method and Class, Empty Class, Data Class • Keyword Arguments	
		Day 4	Hour# 1, 2	• Data and its types (structured, Unstructured) • Quantitative data, Numerical, Continuous, and Discrete variables	

	Overview			• Qualitative data, Categorical, Nominal, Ordinal, and Binary variables	
			Hour # 3-4	Measures of Central Tendency •Mean, Mode, Median	
		Day 5	Hour# 1,2	Measures of Dispersion • Variance, Standard deviation • Co-efficient of variation, skewness and kurtosis	
			Hour# 3, 4	Measures of Position •Z-Score, Percentile, Quartile	
Week 3	Descriptive Statistics and Probability Overview	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	• Task 28-48 *Details may be seen at Annexure-I*
			Hour# 2	• Correlation Coefficient	
			Hour# 3	• Univariate, bivariate and multivariate plots	
			Hour# 4	Probability	
		Day 2	Hour# 1	Joint, Marginal and Conditional probability	
			Hour# 2	• Probability Distributions •	
			Hour # 3-4	• Discrete and Continuous probability distributions • Bayesian Probability	
	Python Support Libraries for Exploratory Data Analysis - NUMPY	Day 3	Hour# 1	• Introduction to Numpy	
			Hour# 2,3,4	• Creating Numpy Arrays (from Python list, from built-in methods, from random) • Array Attributes and Methods (reshape, max, min, argmax, argmin, shape, dtype, size, ndim) • Operations on Arrays (copying, append and Insert, Sorting, Removing/Deleting, Combining/Concatenating, Splitting)	
		Day 4	Hour # 1-2	• Data Loading & Saving • NumPy Indexing and Selection (Indexing a 2D array, Logical Selection) • Broadcasting	
			Hour # 3-4	• Type Casting • Arthmetic Operations (Add, Subtract, Multiply, Divide, Exponentiation) • Universal Array Functions (sqrt, exp, max,	

				sin, etc)	
	- Pandas	Day 5	Hour# 1	• Introduction to Pandas	
			Hour# 2	• Series and DataFrame and Data Input • Selection and Indexing (rows, columns, conditional selection, selection of subset of rows and columns, index setting, etc)	
			Hour# 3	• Operations on DataFrames (head, unique, value counts, applying custom functions, getting column and index names, sorting and ordering, null value check, value replacement, dropping rows and columns, etc)	
			Hour# 4	• Missing data & its handling	
Week 4	Python Support Libraries for Exploratory Data Analysis - Pandas - Seaborn	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	• Task 28-48 *Details may be seen at Annexure-I*
			Hour# 2	• Merging, Joining, and Concatenation (inner, outer, right and left joins)	
			Hour # 3-4	• GroupBy • Discretization and Binning • Operations on DataFrames • Data output/saving • Pandas for Plotting (area, bar, density, hist, line, scatter, barh, box, hexbin, kde, and pie plots	
		Day 2	Hour# 1	• Introduction to Seaborn	
			Hour# 2	Distribution Plots •distplot •jointplot (pairplot, rugplot, kdeplot)	
			Hour# 3	Categorical Data Plots •factorplot, boxplot, violinplot, stripplot, swarmplot, barplot, countplot	
			Hour# 4	Matrix Plots •Heatmap	
		Day 3	Hour# 1	• Machine learning introduction and types	
			Hour# 2,3,4	• Classical machine learning pipeline (data collection, preprocessing, feature crafting, modeling, testing and evaluation, and deployment)	
		Day 4	Hour # 1,2	• Supervised machine learning • Regression and classification problems • Components of supervised machine learning (labeled data, hypothesis, cost function, optimizer)	
			Hour # 3,4	• Univariate Linear Regression with Gradient Descent	

		Day 5	Hour # 1-2	• Univariate Linear Regression with Gradient Descent • Without Vectorization	
			Hour # 3-4	• With Vectorization	
Week 5	Machine Learning-I	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	• Task – 51,52 *Details may be seen at Annexure-I*
			Hour# 2,3,4	• Multivariate Linear Regression	
		Day 2	Hour# 1,2,3, 4	• Polynomial Regression	
		Day 3	Hour# 1,2,3, 4	• Logistic Regression (Binary Classification)	
		Day 4	Hour# 1,2,3, 4	• Logistic Regression (Multiclass Classification)	
		Day 5	Hour# 1,2,3, 4	• Code practice	
Week 6	Natural Language Processing	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	• Task 53-55 *Details may be seen at Annexure-I*
			Hour# 2	• Introduction to Natural Language Processing	
			Hour# 3	• Syntax, Semantics, Pragmatics, and Discourse • NLP curves and future directions	
			Hour# 4	Data pre-processing for NLP •Introduction to NLTK/SpaCy •Noise removal (stopwords, punctuation, etc)	
		Day 2	Hour# 1	• Word and sentence tokenization • Word segmentation • Stemming • Text normalization • Regular expression for string parsing	
			Hour # 2-3	• POS tagging • NER tagging • Chunking and Chinking • Lemmatization • WordNet	
			Hour# 4	• Words as features (BoW model) • Feature Selection and Extraction • Document Similarity	
	Machine Learning II	Day 3	Hour# 1	• Testing	
			Hour# 2	• Evaluation Metrics • Classification and Regression	

Artificial Intelligence Machine Learning
Deep Learning

			Hour # 3-4	• Dataset imbalance and its remedies (Augmentation)	
		Day 4	Hour# 1,2,3	• Support Vector Machine (SVM)	
			Hour# 4	• Decision Tree	
		Day 5	Hour# 1,2	• Decision Tree	
			Hour # 3-4	• Bagging – Random Forest	
Build Your CV – Mid-term Exam					
Week 7	Deep Learning I	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	• Task 56-64 *Details may be seen at Annexure-I*
			Hour# 2,3,4	• Boosting	
		Day 2	Hour# 1,2,3, 4	MLP Feed Forward Neural Network • Forward and backward passes • Nonlinearity: Activation functions • Cross-Entropy • Computational graph and Backpropagation • Vanishing and exploding gradients • Overfitting, underfitting, dropout regularization	
		Day 3	Hour# 1,2,3, 4	• Introduction and implementation of neural networks using appropriate deep learning API of choice (TensorFlow, PyTorch, Keras)	
		Day 4	Hour # 1-2	Convolutional Neural Network (CNN) •2D CNN for image classification	
			Hour # 3-4	• 1D CNN for text document classification	
		Day 5	Hour # 1-2	• Code Practice Neural Networks	
			Hour # 3-4	• Code Practice Neural Networks	
Week 8	Deep Learning II	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	
			Hour# 2,3,4	• Recurrent Neural Networks (RNNs)	
		Day 2	Hour# 1,2,3, 4	• Long-Short-Term-Memory Networks (LSTM)	
		Day 3	Hour# 1	• LSTM Code Practice	
		Day 4	Hour# 1,2,3, 4	• Gated Recurrent Unit Networks	

		Day 5	Hour #1,2,3,4	• GRU Code Practice	
Week 9	Deep Learning II	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	
			Hour # 2,3,4	Word Embeddings • Word2vec • Continuous BOW • Continuous Skip-gram	
		Day 2	Hour# 1,2,3, 4	• Gensim and Custom Embedding Training	
		Day 3	Hour# 1,2,3, 4	• Sequence Models	
		Day 4	Hour# 1,2,3, 4	Sequence Models • 1 to 1 • 1 to Many	
		Day 5	Hour# 1,2,3, 4	Sequence Models •Many to 1 •Many to Many	
Week 10	Deep Learning II	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	• Task 65 *Details may be seen at Annexure-I*
			Hour# 2,3,4	• Bi-Directional LSTM/RNN in Sequence Models	
		Day 2,3	Hour# 1,2,3, 4	• Attention Mechanism in Models	
	Employable Project / Assignment (2 weeks, 11-12) in addition of regular classes. OR On job training (2 weeks)	Day 4,5	Hour# 1,2,3, 4	Selection of Project, architecture discussion, preparation. • Guidelines to the Trainees for selection of employable project like final year project (FYP). • Assignment of Independent project to each Trainee. • A project based on trainee's aptitude and acquired skills. • Designed by keeping in view the emerging trends in the local market as well as across the globe. • The project idea may be based on entrepreneurship. • Leading to the successful employment. • The duration of the project will be 2 weeks • Ideas may be generated via different sites such as: https://1000projects.org/ https://nevonprojects.com/ https://www.freestudentprojects.com/	

				https://technofizi.net/best-computer- science-and-engineering-cse-project- topics-ideas-for-students/ • Final viva/assessment will be conducted on project assignments. • At the end of session, the project will be presented in skills competition. • The skill competition will be conducted on zonal, regional and National level. • The project will be presented in front of Industrialists for commercialization • The best business idea will be placed in NAVTTC business incubation center for commercialization. **OR** • On job training for 2 weeks: • Aims to provide 2 weeks industrial training to the Trainees as part of overall training program • Ideal for the manufacturing trades • As an alternate to the projects that involve expensive equipment • Focuses on increasing Trainee's motivation, productivity, efficiency and quick learning approach.	
Week 11	MS Azure AI Service	Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	• Task 65 *Details may be seen at Annexure-I*
			Hour # 2,3	Selection of Microsoft Azure AI Service •Selection the appropriate service for a vision solution •Selection the appropriate service for a language analysis solution	
			Hour# 4	• Selection the appropriate service for a decision support solution	
		Day 2	Hour # 1,2	• Selection the appropriate service in Cognitive Services for a speech solution • Selection the appropriate Applied AI services	
			Hour # 3-4	• Configuring Security for Microsoft Azure AI Service • Manage account keys • Manage authentication for a resource • Secure services by using Azure Virtual Networks • Plan for a solution that meets Responsible AI principles	
		Day 3	Hour # 1-2	Create & Manage Microsoft Azure AI Service •Create an Azure AI resource •Configure diagnostic logging	

			Hour # 3-4	• Manage costs for Azure AI services • Monitor an Azure AI resource	
		Day 4	Hour # 1-2	• Deploy Microsoft Azure AI Services • Determine a default endpoint for a service • Create a resource by using the Azure portal • Integrate Azure AI services into a continuous integration/continuous deployment (CI/CD) pipeline • Plan a container deployment • Implement prebuilt containers in a connected environment	
			Hour # 3-4	Microsoft Azure Creation of Solutions for Anomaly Detection Content Improvement •Create a solution that uses Anomaly Detector, part of Cognitive Services •Create a solution that uses Azure Content Moderator •Create a solution that uses Personalizer, part of Cognitive Services •Create a solution that uses Azure Metrics Advisor, part of Azure Applied AI Services •Create a solution that uses Azure Immersive Reader, part of Azure Applied AI Services	
		Day 5	Hour # 1-2	Microsoft Azure Implementation of Image and Video Processing Solutions •Analyze images •Extract text from images	
			Hour # 3-4	• Implement image classification and object detection by using the Custom Vision service, part of Azure Cognitive Services	
Week 12		Day 1	Hour# 1	• Motivational Lecture (For further detail please see Page No: 3& 4)	• Task 65 *Details may be seen at Annexure-I*
			Hour# 2,3,4	• Process videos	
		Day 2	Hour# 1,2,3, 4	Microsoft Azure Natural Language Processing (NLP) Solutions Implementation •Analyze text •Process speech •Translate language	
		Day 3	Hour# 1,2,3, 4	• Build and manage a language understanding model • Create a question answering solution	
		Day 4	Hour# 1	• Build and manage a language understanding model	
			Hour # 2-4	Microsoft Azure Knowledge Mining Solutions Implementation	
		Day 5	Hour # 1-4	Microsoft Azure Conversational AI Solutions Implementation	

Annexure-I

List of Tasks

Task No.	Task Title	Description	Week
1.	Installation	Download and install Anaconda3 Install PyTorch Install TensorFlow 2.0 Install VSCode Install PyCharm	1
2.	Linux Commands	Practice these commands: pwd, cd, ls, cat, sudo, man, redirection, mkdir, rm, rmdir, cp, mv, file, reading, cat, more, less, head, alias, shutdown, restart, touch, nano, bash, sh, chmod, ps, kill, dpkg	1

3.	Python	```python
		# This program adds two numbers
		
		num1 = 1.5
		num2 = 6.3
		
		# Add two numbers
		sum = num1 + num2
		
		# Display the sum
		print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))
		``
		1	

4.	Python	
		`# Store input numbers`
		`num1 = input('Enter first number: ')`
		`num2 = input('Enter second number: ')`
		
		`# Add two numbers`
		`sum = float(num1) + float(num2)`
		
		`# Display the sum`
		`print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))`
			1

5.	Python	
		`# Python Program to calculate the square root`
		
		`# Note: change this value for a different result`
		`num = 8`
		
		`# To take the input from the user`
		`#num = float(input('Enter a number: '))`
		
		`num_sqrt = num ** 0.5`
		`print('The square root of %0.3f is %0.3f'%(num ,num_sqrt))`
			1

6.	Python	
		`# Find square root of real or complex numbers`
		`# Importing the complex math module`
		`import cmath`
		
		`num = 1+2j`
		
		`# To take input from the user`
		`#num = eval(input('Enter a number: '))`
		
		`num_sqrt = cmath.sqrt(num)`
		`print('The square root of {0} is {1:0.3f}+{2:0.3f}j'.format(num ,num_sqrt.real,num_sqrt.imag))`
			1

7.	Python	
		`# Python Program to convert temperature in celsius to fahrenheit`
		
		`# change this value for a different result`
		`celsius = 37.5`
		
		`# calculate fahrenheit`
		`fahrenheit = (celsius * 1.8) + 32`
		`print('%0.1f degree Celsius is equal to %0.1f degree Fahrenheit' %(celsius,fahrenheit))`
			1

8.	Python		1

```python
# Python Program to find the area of triangle

a = 5
b = 6
c = 7

# Uncomment below to take inputs from the user
# a = float(input('Enter first side: '))
# b = float(input('Enter second side: '))
# c = float(input('Enter third side: '))

# calculate the semi-perimeter
s = (a + b + c) / 2

# calculate the area
area = (s*(s-a)*(s-b)*(s-c)) ** 0.5
print('The area of the triangle is %0.2f' %area)
```

9.	Python	
		`# Solve the quadratic equation ax**2 + bx + c = 0`
		
		`# import complex math module`
		`import cmath`
		
		`a = 1`
		`b = 5`
		`c = 6`
		
		`# calculate the discriminant`
		`d = (b**2) - (4*a*c)`
		
		`# find two solutions`
		`sol1 = (-b-cmath.sqrt(d))/(2*a)`
		`sol2 = (-b+cmath.sqrt(d))/(2*a)`
		
		`print('The solution are {0} and {1}'.format(sol1,sol2))`
			1

10.	Python	
		`# Taking kilometers input from the user`
		`kilometers = float(input("Enter value in kilometers: "))`
		
		`# conversion factor`
		`conv_fac = 0.621371`
		
		`# calculate miles`
		`miles = kilometers * conv_fac`
			1

11.	Python	`i = 10`
		`if (i > 15):`
		`  print ("10 is less than 15")`
		`print ("I am Not in if")`
			1

12.	Python	<pre>i = 20;
		if (i < 15):
		    print ("i is smaller than 15")
		    print ("i'm in if Block")
		else:
		    print ("i is greater than 15")
		    print ("i'm in else Block")
		print ("i'm not in if and not in else Block")</pre>
			1

13.	Python	<pre>i = 10
		if (i == 10):
		    # First if statement
		    if (i < 15):
		        print ("i is smaller than 15")
		    # Nested - if statement
		    # Will only be executed if statement above
		    # it is true
		    if (i < 12):
		        print ("i is smaller than 12 too")
		    else:
		        print ("i is greater than 15")</pre>
			1

14.	Python	<pre>i = 20
		if (i == 10):
		    print ("i is 10")
		elif (i == 15):
		    print ("i is 15")
		elif (i == 20):
		    print ("i is 20")
		else:
		    print ("i is not present")</pre>
			1

15.	Python	Exercise on for loops in Python: https://www.geeksforgeeks.org/python-for-loops/	1
16.	Python	Exercise on While loops in Python: https://www.geeksforgeeks.org/python-while-loops/	1
17.	Python	Exercise on Break statement in Python: https://www.geeksforgeeks.org/python-break-statement/	1
18.	Python	Exercise on Continue statement in Python: https://www.geeksforgeeks.org/python-continue-statement/	1
19.	Python	Exercise on various looping techniques in Python: https://www.geeksforgeeks.org/looping-techniques-python/	1
20.	Python	Exercise on User defined functions in Python: https://www.geeksforgeeks.org/functions-in-python/	2

21.	Python	Exercise on List data type in Python: https://www.programiz.com/python-programming/list	1
22.	Python	Exercise on Tuple data type in Python: https://www.programiz.com/python-programming/tuple	1
23.	Python	Exercise on String data type in Python: https://www.programiz.com/python-programming/string	1
24.	Python	Exercise on Set data type in Python: https://www.programiz.com/python-programming/set	1
25.	Python	Exercise on Dictionary data type in Python: https://www.programiz.com/python-programming/dictionary	1
26.	Python	Exercise on Exception Handling in Python: https://www.programiz.com/python-programming/exception-handling	2
27.	Python	Exercise on User defined Exception Handling in Python: https://www.programiz.com/python-programming/user-defined-exception	2
28.	Numpy	Exercise on Numpy create Array Using Python: https://www.w3schools.com/python/numpy_creating_arrays.asp	3,4
29.	Numpy	Exercise on Numpy Indexing in Array Using Python: https://www.w3schools.com/python/numpy_array_indexing.asp	3,4
30.	Numpy	Exercise on Numpy Slicing in Array Using Python: https://www.w3schools.com/python/numpy_array_slicing.asp	3,4
31.	Numpy	Exercise on Numpy Slicing in Array Using Python: https://www.w3schools.com/python/numpy_data_types.asp	3,4
32.	Numpy	Exercise on Numpy Array coping and viewing : https://www.w3schools.com/python/numpy_copy_vs_view.asp	3,4
33.	Numpy	Exercise on Numpy Array Shaping : https://www.w3schools.com/python/numpy_array_shape.asp	3,4
34.	Numpy	Exercise on Numpy Array reshaping : https://www.w3schools.com/python/numpy_array_reshape.asp	3,4
35.	Numpy	Exercise on Numpy Array iteration: https://www.w3schools.com/python/numpy_array_iterating.asp	3,4
36.	Numpy	Exercise on Numpy Matrix joining https://www.w3schools.com/python/numpy_array_join Week 4.asp	3,4
37.	Numpy	Exercise on Numpy Array splitting https://www.w3schools.com/python/numpy_array_split.asp	3,4
38.	Numpy	Exercise on Numpy Array searching https://www.w3schools.com/python/numpy_array_search.asp	3,4
39.	Numpy	Exercise on Numpy Array sorting https://www.w3schools.com/python/numpy_array_sort.asp	3,4
40.	Numpy	Exercise on Numpy Array Random technique https://www.w3schools.com/python/numpy_random.asp	3,4
41.	Pandas	Exercise on Pandas basics: https://www.w3schools.com/python/pandas_tutorial.asp	3,4
42.	Pandas	Exercise on Pandas installation: https://www.w3schools.com/python/pandas_getting_started.asp	3,4
43.	Pandas	Exercise on Pandas Series data https://www.w3schools.com/python/pandas_series.asp	3,4
44.	Pandas	Exercise on Pandas Data Frame: https://www.w3schools.com/python/pandas_dataframes.asp	3,4
45.	Pandas	Exercise on Pandas Open CSV files: https://www.w3schools.com/python/pandas_csv.asp	3,4

46.	Pandas	Exercise on Pandas Data analyzation: https://www.w3schools.com/python/pandas_analyzing.asp
47.	Pandas	Exercise on Pandas Data Cleaning techniques: https://www.w3schools.com/python/pandas_cleaning.asp
48.	Pandas	Exercise on Pandas Data Correlation: https://www.w3schools.com/python/pandas_correlations.asp
49.	Stats	Perform Mean, Midian and mode: https://www.w3schools.com/python/python_ml_mean_median_mode.asp
50.	Stats	Perform Standard Deviation: https://www.w3schools.com/python/python_ml_standard_deviation.asp
51.	Machine Learning	Implement Linear Regression https://stackabuse.com/linear-regression-in-python-with-scikit-learn/
52.	Machine Learning	Perform Logistics Regression: https://towardsdatascience.com/logistic-regression-using-python-sklearn-numpy-mnist-handwriting-recognition-matplotlib-a6b31e2b166a https://www.datacamp.com/community/tutorials/understanding-logistic-regression-python
53.	Machine Learning	Exercise on Decision Tree: https://www.datacamp.com/community/tutorials/decision-tree-classification-python
54.	Machine Learning	Exercise on SVM: https://stackabuse.com/implementing-svm-and-kernel-svm-with-pythons-scikit-learn/
55.	Machine Learning	Exercise on Time Series Analysis: https://www.dataquest.io/blog/tutorial-time-series-analysis-with-pandas
56.	Machine Learning	Demonstration of Neural Networks: https://www.analyticsvidhya.com/blog/2020/07/neural-networks-from-scratch-in-python-and-r
57.	Deep Learning	Exercise on MLP: https://machinelearningmastery.com/neural-networks-crash-course/
58.	Deep Learning	Exercise on Feed Forward neural networks: https://builtin.com/data-science/feedforward-neural-network-intro
59.	Deep Learning	Exercise on Neural Network: https://www.analyticsvidhya.com/blog/2019/08/detailed-guide-7-loss-functions-machine-learning-python-code/
60.	Deep Learning	Exercise on Linguistics using Machine learning in python: https://medium.com/towards-artificial-intelligence/natural-language-processing-nlp-with-python-tutorial-for-beginners-1f54e610a1a0
61.	Deep Learning	Text processing: https://pythonspot.com/category/nltk/
62.	Deep Learning	Text Analysis https://medium.com/towards-artificial-intelligence/natural-language-processing-nlp-with-python-tutorial-for-beginners-1f54e610a1a0
63.	Deep Learning	Demonstrate Convolution Neural Network: https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53
64.	Deep Learning	Perform CNN on CIFAR-10 Dataset (https://www.analyticsvidhya.com/blog/2020/02/learn-image-classification-cnn-convolutional-neural-networks-3-datasets/)
65.	Microsoft Azure	Microsoft Azure Video Lectures at Microsoft Learning